Data Checks

Checks Run	Checks With Findings	Checks Fixed	Checks Excluded	Checks Flagged	Row Counts Match
39	21	10	3	8	18 / 18

check_id	table_name	check	rows	decision	action
V01	mes_batches	Exact duplicate batch rows	5	Fix	Removed (Table.Distinct)
V02	mes_batches	batch_id still repeated after removing exact duplicates	0	None needed	
V03	mes_batches	process_order_id not found in erp_process_orders (trailing space, lower case, missing hyphen)	9	Fix	Trim, upper-case, restore PO- prefix; all 669 then join 1:1
V04	mes_batches	fill_end blank	4	Fix	Taken from the FILLING step end in mes_batch_steps
V05	mes_batches	qc_sample_units blank on a commercial batch	14	Fix	Derived from the unit balance: filled - rejected - good
V06	mes_batches	Media fills without inspection, QC or good-unit counts	6	None needed	Expected: media fills are inspected outside MES; quality KPIs exclude them, line-time KPIs keep them
V07	mes_batches	units_good <> units_filled - rejects - QC samples	0	None needed	
V08	mes_batches	units_filled + line_loss_units > theoretical_units	0	None needed	
V09	mes_batches	Product filled on a line it is not approved for	0	None needed	
V10	mes_batches	reinspection_flag disagrees with a RE-INSPECTION step	0	None needed	
V11	mes_batches	Final reject rate above the 2.5% alert limit without re-inspection	0	None needed	Note: final counts are after re-inspection, so they understate first-pass rejects
V12	mes_batch_steps	Commercial batch missing SETUP, FILLING, INSPECTION or PACKAGING	0	None needed	
V13	mes_batch_steps	step_end before step_start	0	None needed	Steps pause over weekends (elapsed time), so the dashboard uses medians
V14	mes_downtime_events	end_time before start_time	5	Fix	Start and end swapped; all 5 then sit inside their batch phase and the shift label fits
V15	mes_downtime_events	Stop ends after its batch's setup or filling step ended (82-137 h)	3	Exclude	Minutes set to blank, stop still counted; historian shows the line running
V16	mes_downtime_events	Double-logged stop: same equipment, start within 5 min of a lower event ID	35	Exclude	Dropped from minutes and counts (all are EV-501825 to EV-501859)

status

Exclude

Fix

Flag

Info

Pass

Limitations and asks for data engineering

- Inspection counts are final, after re-inspection, so the first-pass reject rate cannot be measured. Ask MES to store both counts.
- The true length of the three long stops is unknown. Ask for an end-time check against the running batch.
- QA staffing is not in any table, so the cause of the QA queue (issue 3) is inferred, not proven.
- Only the latest due-date change is kept. Ask ERP to keep the original promise date as its own field.
- The newest batches may still get deviations, and 23 are still in QA review, so the last two weeks of FPQ and yield are provisional.
- Inspection and packaging pause over weekends, so step times are elapsed hours. The dashboard uses medians for that reason.
- There are no targets for October 2026, and the data stops on 1 Oct 2026.

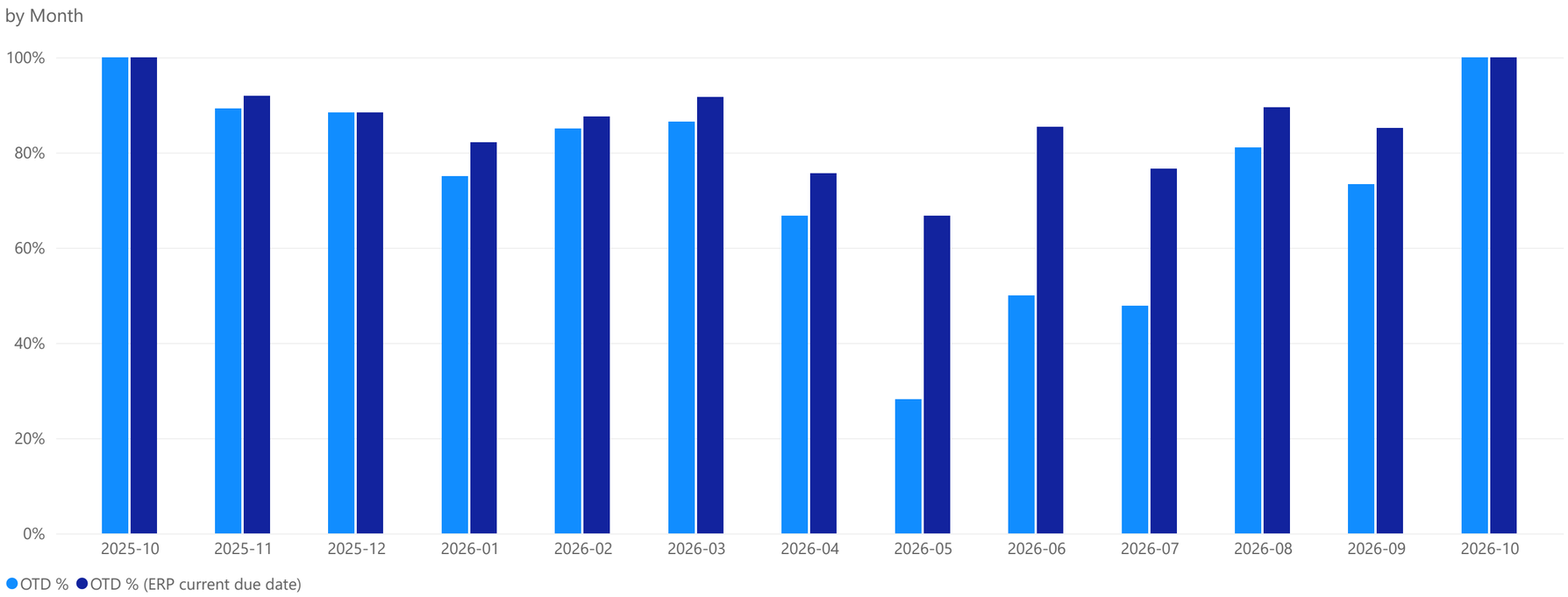
table_name	actual_rows	expected_rows	matches_dictionary
cmms_equipment	21	21	True
cmms_work_orders	273	273	True
erp_clients	5	5	True
erp_process_orders	669	669	True
erp_products	11	11	True
erp_sales_orders	488	488	True
erp_shipments	637	637	True
erp_suppliers	6	6	True
historian_line_hourly	26489	26489	True
kpi_targets	180	180	True
mes_batch_steps	2721	2721	True
mes_batches	680	680	True
mes_component_usage	1798	1798	True
mes_downtime_events	1859	1859	True
mes_inspection_rejects	4014	4014	True

r	pi	formula	definition	dax_measure	includes	excludes	unit	date_attribution
1	Throughput	SUM(units_good) / SUM(scheduled_hours x weekly target / standard week hours)	Good units filled, against a target pro-rated for the hours each line was scheduled.	[Good Units], [Throughput Target Units], [Throughput vs Target %]	Commercial batches; holidays and shutdowns lower the target	Media fills; the 5 duplicate MES rows	units per week; %	Fill-end date of the batch
2	Yield	SUM(units_good, 0 if QA rejected the batch) / SUM(theoretical_units)	Share of the theoretical units that became released, saleable units.	[Yield %]	Commercial batches; batches still in QA review count with MES good units (provisional)	Media fills; duplicate MES rows	%	Fill-end date of the batch
3	First-pass quality	COUNT(batches with reinspection_flag = N and no Major/Critical deviation) / COUNT(batches)	Share of batches that needed no re-inspection and had no major or critical deviation.	[FPQ %]	Commercial batches; minor deviations do not fail a batch	Media fills	% of batches	Fill-end date of the batch
4	Scrap rate	SUM(line_loss + rejects + good units of rejected batches) / SUM(theoretical_units); EUR = scrap units x std cost	Units scrapped at the filler, at inspection and in QA-rejected batches, valued at standard cost.	[Scrap %], [Scrap EUR]	Commercial batches; rejected batches in full	QC samples (planned use); unfilled bulk (shown in yield); media fills	%; EUR	Fill-end date of the batch
5	Unplanned downtime	SUM(unplanned stop minutes, phase = Filling) / SUM(filling minutes)	Unplanned stop time while filling, as a share of filling time.	[Unplanned Downtime %]	All unplanned reason codes incl. not coded; media fill batches (they use line time)	Planned interventions (PLN-01); setup-phase waits (shown as setup time); 35 duplicate events; minutes of 3 events that end after their batch's filling ended	% of filling time	Fill-end date of the batch the s belongs to
6	On-time delivery	COUNT(orders with all batches shipped <= committed date) / COUNT(orders with committed date in period and before the extract)	Share of orders due in the period that shipped complete on or before the committed date.	[OTD %]	Orders due before 5 Oct 2026, shipped or not; short-shipped orders count as late	Orders not yet due at the extract	% of orders	Committed date (first confirm the new date when the client a the change)

kpi	definition_a_chosen	Conflict Value A	definition_b	Conflict Value B	why_a
Yield	Released good units / theoretical units	95.5%	MES good units / filled units	97.3%	The bulk belongs to the client: losses before filling and rejected batches are real yield losses. The two give opposite answers against 96.5%, so leadership must confirm the target's basis.
First-pass quality	No re-inspection and no major or critical deviation	89.2%	No re-inspection and no deviation of any severity	71.6%	Minor deviations (mostly documentation errors) do not change the batch's route or hold release. Only A fits the 92% target in a normal month (92-95% Oct-Feb).
Scrap	(Line loss + inspection rejects + rejected batches) / theoretical	2.19%	Every unit not released or sampled / theoretical (adds unfilled bulk)	3.20%	B adds about 1% of bulk that never became a unit and is already counted in yield. A matches the 2.0-2.5% targets.
Unplanned downtime	Unplanned stop time during filling / filling time	10.7%	Unplanned stop time / scheduled time	3.8%	Only about a quarter of scheduled time is filling; the rest is setup or idle, where the filler cannot break down. B dilutes failures so every line looks fine.
Throughput	Good units by fill-end week vs target pro-rated by scheduled hours	97.6%	Units released by QA per week vs the same target	93.9%	The weekly view is about what the lines made. B moves output into the week QA released it and swings with the QA queue (19 of 51 weeks differ by more than 25 points).
On-time delivery	Shipped complete by the committed date (first confirmed, or the client's new date)	71.2%	Shipped complete by the ERP current due date	84.5%	62 due dates were moved after the original date had passed (capacity, QA delay, no reason). Measuring against the moved date hides 61 late orders.

line_id

- [ ] FL-1
- [ ] FL-2
- [ ] FL-3



Weekly ops

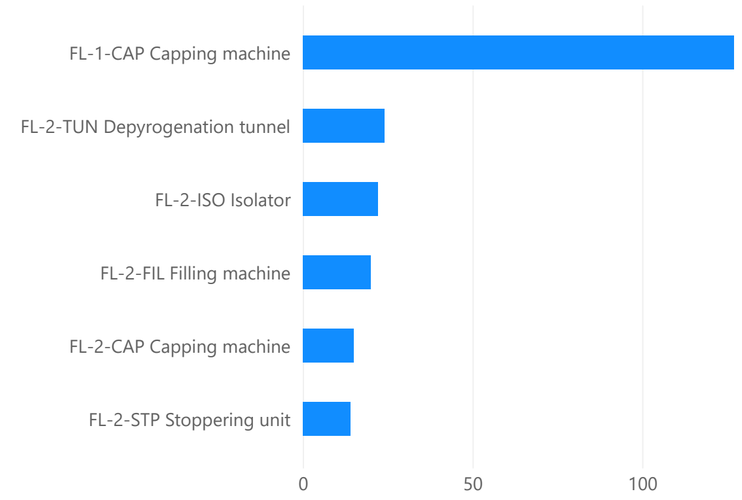
WeekLabel: 2026-W39 (21 Sep)

line_id: FL-1, FL-3, FL-2

kpi	KPI Actual (text)	KPI Target (text)	KPI Gap (text)
On-time delivery	71.4%	96.0%	-24.57 pp
Unplanned downtime	6.0%	9.5%	-3.43 pp
Scrap rate	1.71%	2.18%	-0.47 pp
First-pass quality	100.0%	92.0%	+8.00 pp
Yield	95.8%	96.5%	-0.70 pp
Throughput (good units)	412,905	550,000	-24.9%

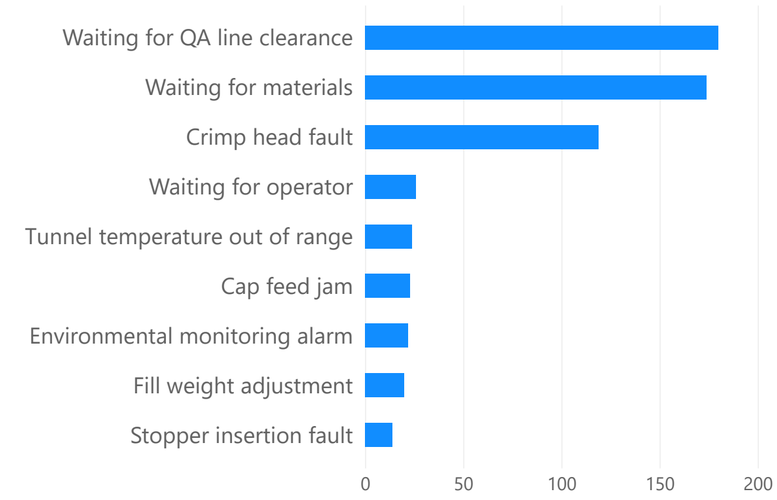
Unplanned stop minutes whilefilling, by equipment

Unplanned Stop Minutes (Filling) and Stops by equipment_label



Unplanned stop minutes by cause (setup and filling)

Unplanned Stop Minutes and First phase by reason_text

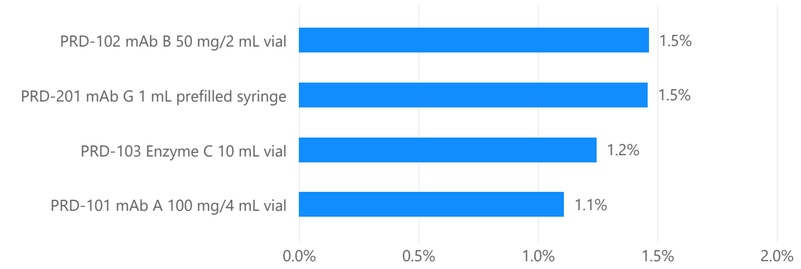


shift_at_ Day Evening Night
fill_start

line_id	Reject Rate %	FPQ %	Reject Rate %	FPQ %	Reject Rate %	FPQ %
FL-1	1.0%	100.0%	1.1%	100.0%	1.4%	100.0%
FL-2	1.3%	100.0%			1.6%	100.0%
FL-3			1.5%	100.0%	1.4%	100.0%

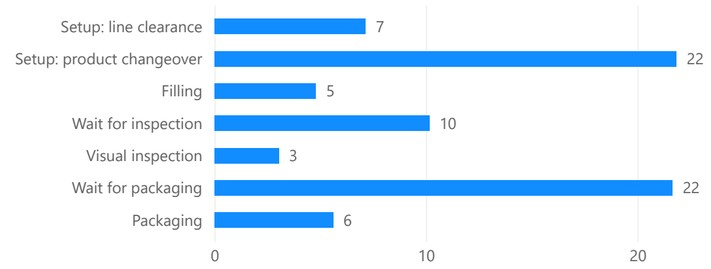
Reject Rate %, Batches and Re-inspected Batches

by product_label



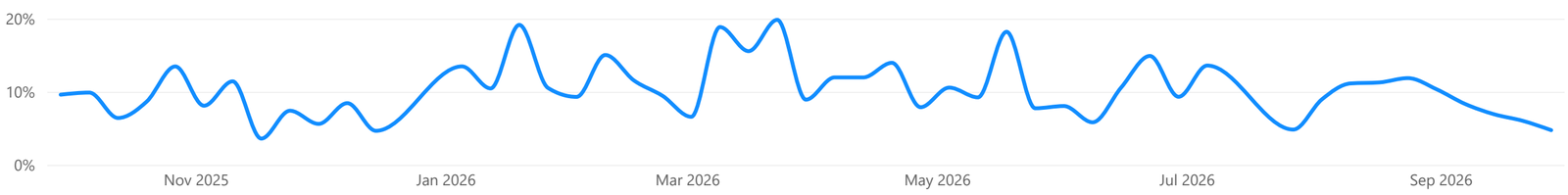
Median Step Hours

by step_name



Unplanned Downtime %

by WeekStart



QA Queue Days (median)

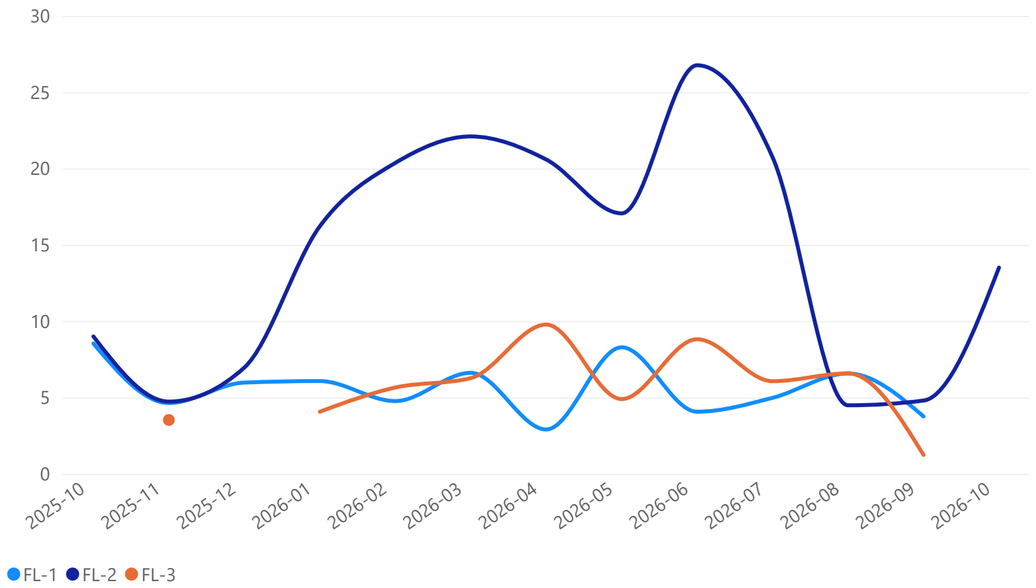
5.00

QA Review Days (median)

4.50

Stops per 100 Filling Hours

by Month and line_id

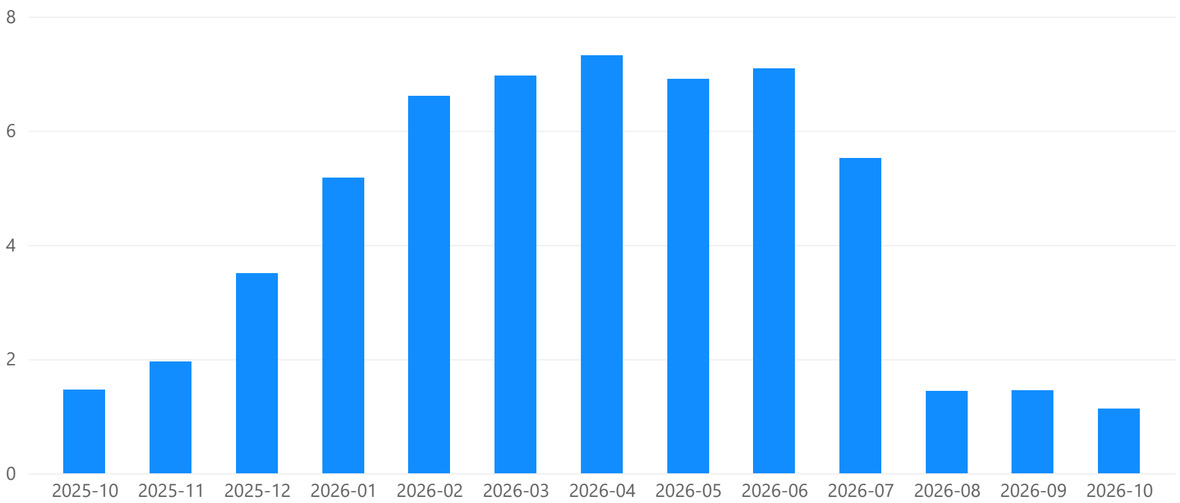


FL-2 stoppering unit: 9 PMs in a row late or cancelled (Nov 2025 to Jul 2026)

work_order_id	wo_type	scheduled_date	completed_date	days_late	status	deferral_reason
WO-610017	PM	Wednesday, October 22, 2025	Thursday, October 23, 2025	1	Completed	
WO-610037	PM	Wednesday, November 19, 2025	Tuesday, January 06, 2026	48	Completed late	Line not available - production priority
WO-610053	PM	Wednesday, December 17, 2025			Cancelled	Line not available - production priority
WO-610082	PM	Wednesday, January 14, 2026			Cancelled	Line not available - production priority
WO-610103	PM	Wednesday, February 11, 2026	Tuesday, March 31, 2026	48	Completed late	Line not available - production priority
WO-610123	PM	Wednesday, March 11, 2026	Wednesday, April 08, 2026	28	Completed late	Line not available - production priority
WO-610143	PM	Wednesday, April 08, 2026	Monday, May 25, 2026	47	Completed late	Line not available - production priority
WO-610163	PM	Wednesday, May 06, 2026			Cancelled	Line not available - production priority
WO-610182	PM	Wednesday, June 03, 2026	Wednesday, July 01, 2026	28	Completed late	Line not available - production priority
WO-610201	PM	Wednesday, July 01, 2026	Saturday, August 01, 2026	31	Completed late	Line not available - production priority
WO-610217	Shutdown	Wednesday, July 15, 2026	Friday, July 17, 2026	2	Completed	
WO-610231	PM	Wednesday, July 29, 2026	Friday, July 31, 2026	2	Completed	
WO-610254	PM	Wednesday, August 26, 2026	Friday, August 28, 2026	2	Completed	
WO-610272	PM	Wednesday, September 23, 2026	Thursday, September 24, 2026	1	Completed	

Alarms per Running Hour

by Month



Corrective Work Orders

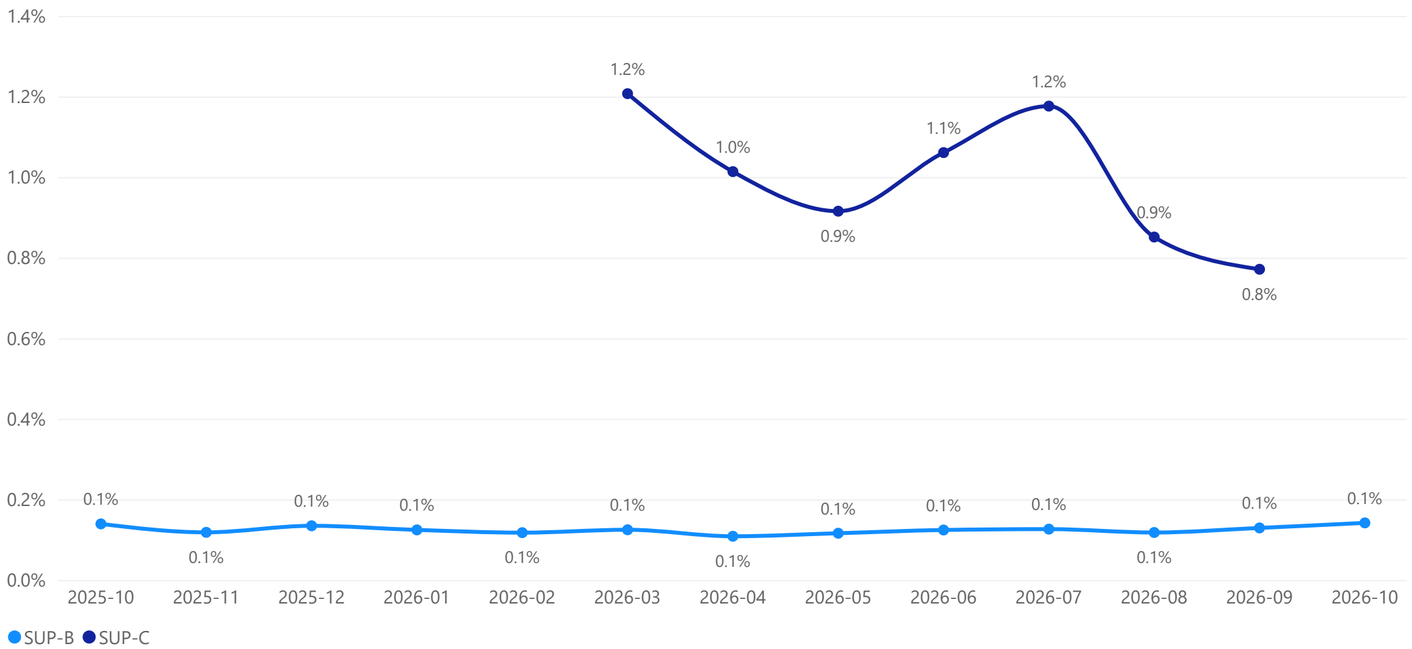
23

PM Compliance %

30.8%

Supplier C plunger stoppers are rejected eight times as often as Supplier B's

Defect Rate % by Month and closure_supplier_id



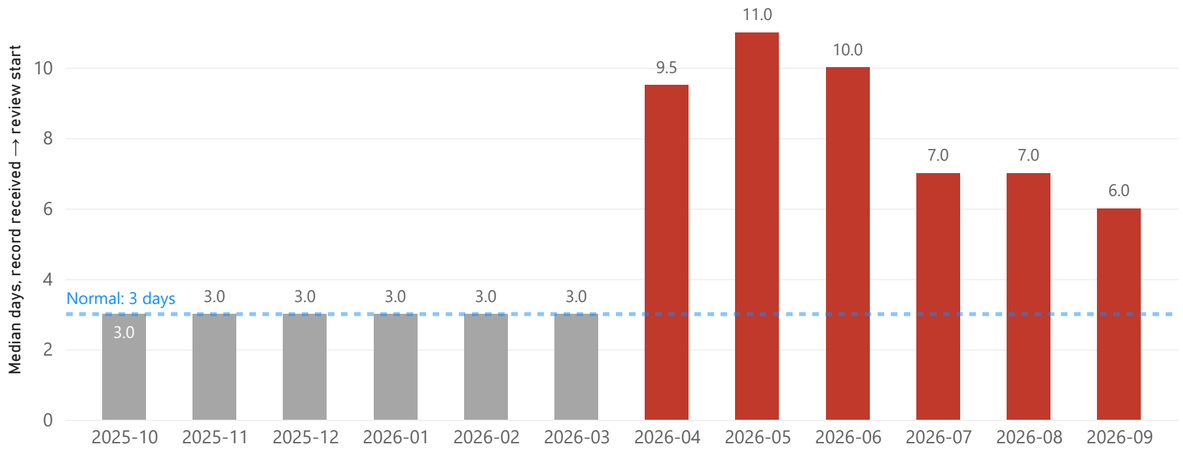
Every Supplier C lot is worse than every Supplier B lot

closure_lot	closure_supplier_id	Batches	Defect Rate %	Re-inspected Batches
CL00817	SUP-C	4	1.9%	4
CL00805	SUP-C	5	1.6%	5
CL00815	SUP-C	3	1.6%	3
CL00808	SUP-C	3	1.5%	3
CL00806	SUP-C	5	1.4%	5
CL00818	SUP-C	6	1.2%	1
CL00811	SUP-C	7	1.2%	6
CL00813	SUP-C	5	1.1%	2
CL00814	SUP-C	3	1.0%	3
CL00816	SUP-C	5	1.0%	1
CL00821	SUP-C	5	1.0%	
CL00809	SUP-C	4	1.0%	
CL00820	SUP-C	3	0.9%	
CL00810	SUP-C	7	0.7%	
CL00812	SUP-C	5	0.6%	
CL00823	SUP-C	4	0.6%	
CL00807	SUP-C	7	0.6%	
Total		**209**	**0.5%**	**33**

Issue 3: From April the QA review queue tripled and drove most late orders

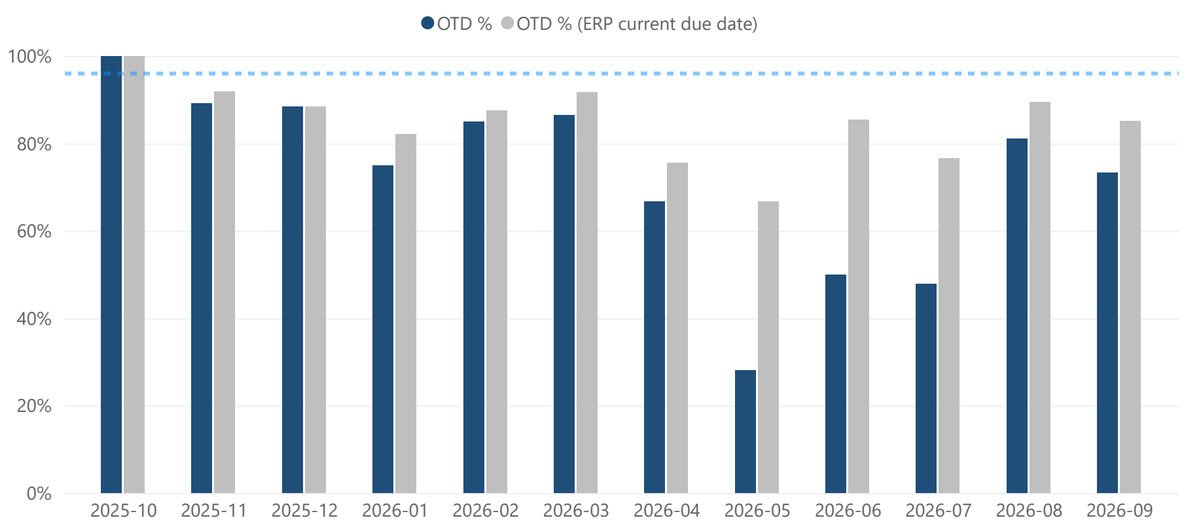
From April, batch records waited three times longer before QA review started

By month the batch finished filling · QMS batch release



OTD % and OTD % (ERP current due date) by Month

On-time delivery fell to 28–67% in April–July; ERP's moved due dates showed 67–85%



132 orders late against the committed date. 34 were re-dated 'QA release delay' after the date had passed

Order	Status	Committed	Shipped	Days Late	ERP re-date reason
SO-70293	Shipped	Wednesday, June 03, 2026	Wednesday, June 10, 2026	7	
SO-70294	Shipped	Tuesday, May 26, 2026	Monday, June 01, 2026	6	
SO-70354	Shipped	Monday, June 22, 2026	Monday, July 06, 2026	14	QA release delay
SO-70276	Shipped	Tuesday, May 26, 2026	Wednesday, June 03, 2026	8	QA release delay
SO-70350	Shipped	Monday, June 22, 2026	Friday, July 03, 2026	11	
SO-70259	Shipped	Monday, April 27, 2026	Tuesday, May 05, 2026	8	
SO-70267	Shipped	Thursday, June 04, 2026	Thursday, June 11, 2026	7	
SO-70269	Shipped	Tuesday, May 26, 2026	Tuesday, June 02, 2026	7	QA release delay
SO-70274	Shipped	Thursday, June 25, 2026	Monday, June 22, 2026	-3	Customer request
SO-70345	Shipped	Monday, July 06, 2026	Tuesday, July 14, 2026	8	
SO-70297	Shipped	Tuesday, June 23, 2026	Friday, June 26, 2026	3	Capacity
SO-70314	Shipped	Monday, July 06, 2026	Wednesday, July 08, 2026	2	
SO-70329	Shipped	Tuesday, June 16, 2026	Tuesday, June 23, 2026	7	
SO-70330	Shipped	Tuesday, July 07, 2026	Friday, July 10, 2026	3	Capacity
SO-70239	Shipped	Wednesday, May 27, 2026	Friday, May 29, 2026	2	
SO-70295	Shipped	Wednesday, June 03, 2026	Wednesday, June 10, 2026	7	
SO-70356	Shipped	Tuesday, August 25, 2026	Thursday, August 20, 2026	-5	Customer request
SO-70208	Shipped	Wednesday, April 22, 2026	Wednesday, April 22, 2026	0	Capacity

Evidence. QMS: the median wait from record received to review start was 3 days every month from October to March, then 9.5–11 days in April–June and 6–7 after. Once a review started, it still took 6 days. ERP orders and shipments: on-time delivery fell to 67%, 28%, 50% and 48% in April–July, and for 79 of the 132 late orders the QA wait was the biggest delay. ERP due-date history: 34 orders were re-dated "QA release delay", all after the original date had passed, so ERP showed only 67–85%.

Impact. 79 late orders, and on average about €13.6M more finished product waiting in the QA queue (April–September).

Ruled out. Volume (records per month were flat: 62, 58, 50 in March–May), deviations (batches without one waited as long), harder reviews (review time unchanged), late records (they reach QA 2 days after filling all year) and one line (all three waited equally).

Not in the data. QA staffing. Confirm it with the QA rota before calling it the cause.